Week 4
Keeping Score
Level 1: Introduction

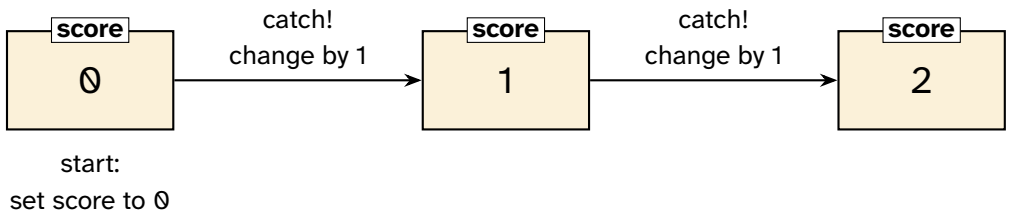
Variables are labeled boxes that remember things.

What you will learn

- How a **variable** remembers a score or lives.
- The difference between **set** and **change**.
- How **pick random** makes your game full of surprises.
- How to build a catch game, and how to give **credit** for art you use.

The big idea

A **variable** is a box with a label. It holds **one** value at a time.

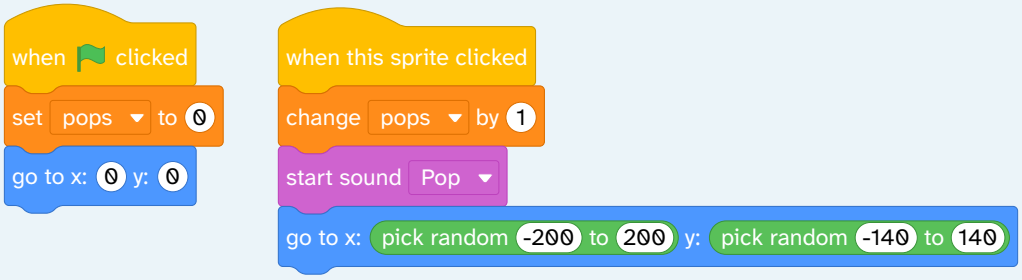


- **set score to 0** empties the box and puts in 0. Use it once, at the start.
- **change score by 1** adds 1 to what is already in the box. Use it during the game.
- **pick random 1 to 6** is like rolling a die: you get a number, but you can't know which!

To make a variable: click **Variables** → **Make a Variable**, type a name, keep "For all sprites", click **OK**. A little score box appears on the stage.

Exercise 4.1: Pop the Balloon

1. New project. Delete the cat. **Choose a Sprite**: pick **Balloon1**.
2. Make a variable called pops.
3. Build both scripts. "pick random" is in **Operators**: drop it into the x and y slots.
4. Click the flag and pop the balloon! Then click the flag again: does pops start at 0?

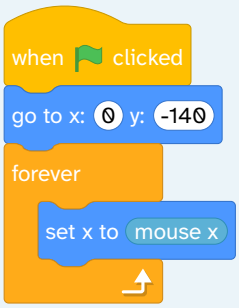


What happens if you take away “set pops to 0”?

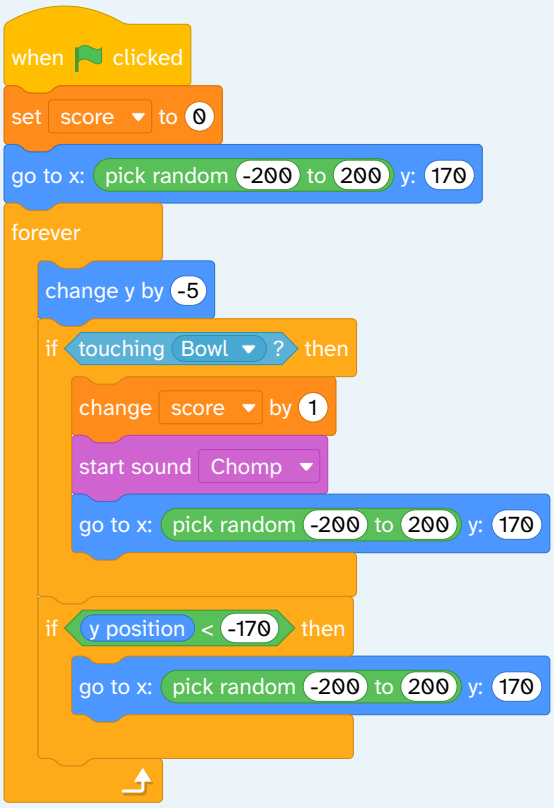
Exercise 4.2: Apple Catch

Catch the falling apples in a bowl!

1. New project. Delete the cat. Add the **Bowl** and the **Apple** sprites. Choose a backdrop you like.
2. Make a variable called `score` (for all sprites).
3. Click the **Bowl** and build its script. “mouse x” is in **Sensing**.
4. Click the **Apple** and build its script. “y position” is in **Motion**; the “<” block is in **Operators**.
5. Play! Move the mouse left and right to catch apples.



Bowl



Apple (pick any sound in the dropdown)

Bug hunt! Mei's apple script is the same, **but** it has no "go to x: (pick random) y: 170" inside the "touching Bowl?" block. Try it: take that block out. What happens to the score when you catch one apple? Why?

Put the block back. **Done when:** each apple you catch adds exactly 1, and missed apples start again at the top.

Exercise 4.3: Lives and Game Over (bonus)

1. Make a variable `lives`. In the Apple's script, under "set score to 0", add "set lives to 3".
2. Inside the "y position < -170" block (a missed apple), add "change lives by -1".
3. At the bottom of the forever loop, add this:



Done when: you start with 3 lives, lose one for each missed apple, and the game stops at 0.

Digital citizenship & safety: Give credit

The Apple, Bowl and Balloon were drawn by real people and shared in the Scratch library for everyone to use. Add a credits note: right-click the code area → **Add Comment**, then type: "Credits: sprites and sounds from the Scratch library." If you change a friend's project, that is a **remix**: say whose project you started from.

Vocabulary box

variable A labeled box that remembers one value, like a score.

set Put a new value in the box (the old one is thrown away).

change Add to the value already in the box (use a minus number to take away).

random A surprise number you cannot predict, like rolling a die.

credit Saying who made the art, sounds or ideas you used.

remix Your own new version of someone else's project, with credit.

Reflection

What is the difference between "set" and "change"?

Whose art did you use today? How did you give credit?

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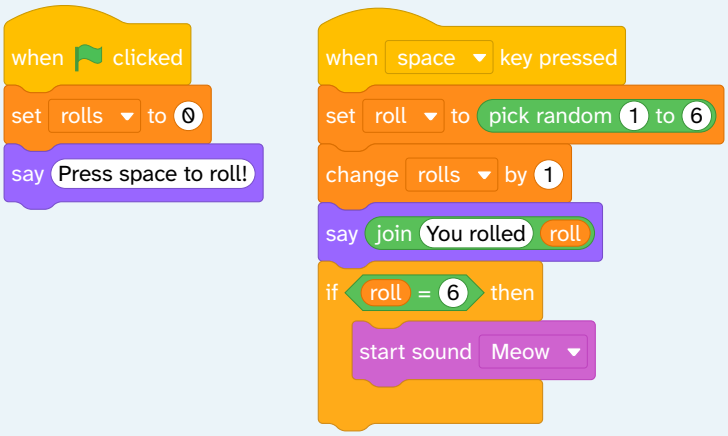
15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Magic Dice (Scratch, about 10 minutes)

Make a dice roller that remembers how many times you rolled.

1. Open Scratch. Make two variables (**Variables** → **Make a Variable**): `roll` and `rolls`.
2. Build the two scripts. "join" and "pick random" are in **Operators**. Type a space after "You rolled" so the words don't squash together.
3. Click the flag, then press `space` to roll.
4. Save as `magic-dice`.



Roll 10 times. Write your numbers: _____

Could you predict the next number? _____ What does `rolls` show now? _____

★★ Challenge (want more?): Doubles!

Add a second variable `roll2` and set it to "pick random 1 to 6" too. Make the sprite say "Doubles!" when `roll` = `roll2`. Roll until you get doubles. How many rolls did it take? _____

Bug check: Zoe put "set rolls to 0" inside the "when space key pressed" script. What will `rolls` always show? Why?

_____

Creative (make it yours): Real dice faces

Click the **Costumes** tab and paint 6 costumes, one for each dice face, with 1 to 6 dots (dots, not colors, show the number). Put them in order. Then add “switch costume to (roll)” to your space-key script so the dice shows the face you rolled. Credit yourself: add a comment “Dice faces drawn by” + your nickname.

Self-check: tick each box when it is true

- ☐ I made variables and chose “For all sprites”.
- ☐ `rolls` starts at `0` every time I click the green flag.
- ☐ `rolls` goes up by exactly 1 for each roll.
- ☐ I can explain the difference between “set” and “change”.